CNCA Centro Nacional de Computação Avançada

DEUCALION

Good practices @Deucalion

B. Malaca, Rui Silva

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PRR Plano de Recuperação e Resiliência
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Context of Deucalion

- **Investment of 20M€**
- **65/35 Portugal/EU budget**
- **Capable of 10 PFlops**
- **Located in Guimarães**
- **Integrated in EuroHPC**



About Deucalion

ARM partition

1632 nodes

A64FX chip (Fujitsu)

32 GB High bandwidth HBM2 RAM (50% faster)

Access to optimised software



X86 partitions

500 nodes + 33 GPU nodes

2 x AMD EPYC 7742 per node (128 cores)

256 GB RAM

Access to optimised software

ARM partition is in the top 350 supercomputers in the world

Usage overview

In numbers

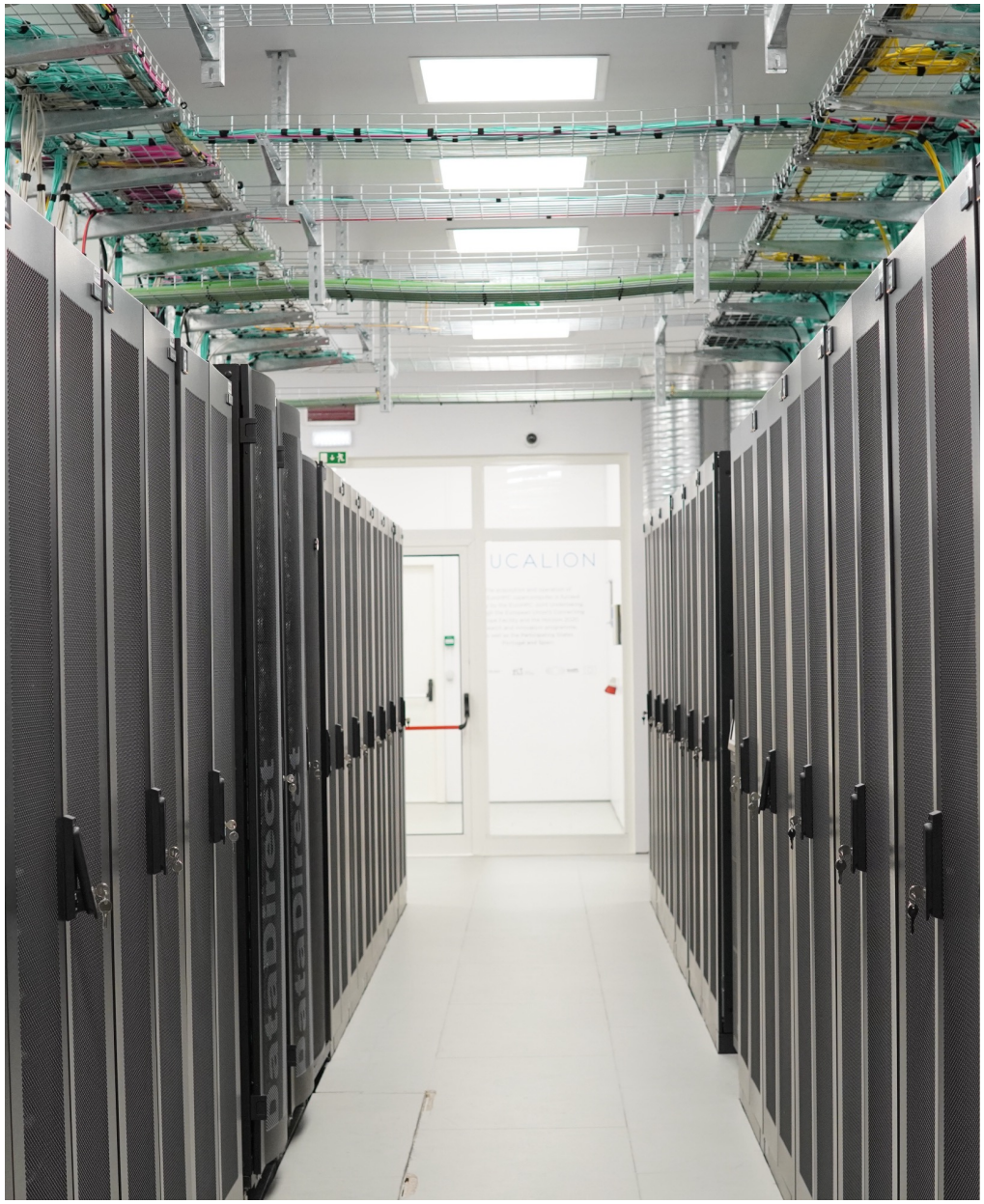
>1500 active users (from Portuguese and European institutions)

>340 active projects

Full x86 and GPU partitions

ARM partition not at full capacity

You will be using the ARM partition for this class



Deucalion structure



x86 Partition
2.3 Pflops
500 nós



Gpus Partition
2.6 Pflops
33 nós com 4 gpus



Arm Partition
5Pflops
1632 nós



Infiniband A Switches

Ethernet Switches

Infiniband B Switches

NVM Storage 430TB

PFS Storage 10PB

NAS 50TB

Login Nodes

Network

Node naming

Login nodes	Compute nodes x86	Compute nodes x86 + GPU	Compute nodes ARM
ln[01-04]	cnx[001-500]	gnx[501-533]	cna[0001-1632]

Aside from the ARM nodes, all nodes have x86 architecture

```
user@localpc ~ % ssh deucalion
Enter passphrase for key '/Users/user/.ssh/deucalion':
(...)

[user@ln01 ~]$ srun -A <account> --time=48:00:00 --
nodes=1  -p normal-arm --pty bash

[user@cna0030 ~]$
```

Some warnings

1500 active users must share 4 login nodes

What happens if all of them suddenly try to compile a program using e.g. GCC?

Possible issues

Abuse of the login nodes: other sessions get affected; users can't login anymore; system gets sluggish.

Abuse of the home folder: other sessions get affected (remember that a lot of configuration files are at your home); all jobs running can be affected if dependent on files at home.

Because of the total number of users in this class, anyone found transgressing any of the rules will see their access to Deucalion revoked

Accessing Deucalion

SSH key

Generate key:

```
ssh-keygen -t ed25519
ENTER
<password>
<repeat password>
```

Upload key:

Go to portal > upper right corner > settings > security

Enter Deucalion

ssh command :

```
ssh -i ~/.ssh/id_ed25519
<username>@ln01.deucalion.macc.fccn.pt
```

From browser (need TOTP):

```
https://login.deucalion.macc.fccn.pt
```

Make your life easier

Edit ~/.ssh/config file:

```
Host deucalion
   HostName
login.deucalion.macc.fccn.pt
   User <username>
   IdentityFile ~/.ssh/id_ed25519
```

ssh command :

```
ssh deucalion
```

Good practices

Home folder

Do not use the home folder extensively. The limits are small (25k files max.).

Create a personal/group folder inside the projects folder.:

```
mkdir /projects/F202500010HPCVLABUMINHO/2026-2027cp/user
```

Accounts

Accounts that end with A/X/G are to use with ARM/x86/GPU nodes.

e.g. f202500010hpcvlabuminhoa is to be used with arm nodes, meaning only using the “dev-arm”, “normal-arm” and “large-arm” partitions

Use the command: `billing`

Limits

Everyone is sharing a certain amount of computing time. Use for what you need but do not overuse it if you are not getting a benefit.

If you ask for an interactive job you are still getting billed 48 cores (regardless if you are using them or not) and we know at all times what’s your personal usage of the machine.

Respect the time given to you (it’s not free) and respect everyone else

Modules

The software in Deucalion is given by modules (just as in the search cluster)

When you load a module in an ARM machine, you will load the ARM-compiled software, so you don't need to worry about mismatches

Loading the same module in the login node will lead to a different binary

```
[user@ln01 ~]$ ml GCC
GCC              GCC/11.3.0      GCC/12.3.0      GCC/13.3.0
GCC/14.3.0       GCC/8.3.0       GCC/10.3.0      GCC/12.2.0
GCC/13.2.0       GCC/14.2.0      GCC/15.2.0
…
[user@ln01 ~]$ ml GCC #load the latest version
```

Jobs

Usually when using a supercomputer you interact via batch jobs (submit a file that has all informations and only use the time you need)

Deucalion also allows to create interactive sessions, but they are not recommended (time wasting)

Useful commands

Get your account:

```
billing #run this in a login node
```

To start a session in an ARM node (max 2 days, 128 nodes):

```
srun -A <account> --time=48:00:00 --nodes=1  -p normal-arm --pty bash
```

Modules:

```
ml av #alias of module avail (does not work in the search cluster)
ml av GCC #will list all modules with GCC in their name
ml GCC #loads latest GCC module
ml GCC/12.3.0 #loads specific version of GCC module
ml purge #unload all modules
```

Practical part

Go to your projects folder:

```
quotaprojects
```

Copy to your space:

```
cp -r /projects/F202500010HPCVLABUMINHO/2026-2027cp/PLprojeto1/ ~
```

Inspect files, e.g.:

```
cd ~/PLprojeto1/
cat MDseqClaudeAI.cpp #or vi MDseqClaudeAI.cpp
```

Useful commands SLURM

Submit job:

```
sbatch <jobscript> #usually .sh file
```

Check job:

```
squeue --me
```

Cancel job:

```
scancel <jobid>
```

Generating batch file

Example (file batchscript.sh):

```
#!/bin/bash
#SBATCH -A f202500010hpcvlabuminhoa
#SBATCH -p normal-arm
#SBATCH -t 10:00 #10 minutes


#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=48


#SBATCH --output=out.o%j
#SBATCH --error=err.e%j

(…)
```